Quantum Denoising for BCI: A Study

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Quantum Denoising: Groundbreaking Theory, Disappointing Results - See the Data!

Introduction

Quantum variational autoencoders (QVAEs, [1]) extract meaningful information into a smaller space, known as latent space. The process is similar to taking notes: the sentence 'My beloved sweater is blue' becomes 'sweater blue.' This is a powerful approach that has **not yet been applied in EEG**-based brain-computer interfaces.

This study hypothesizes that the latent space of a quantum autoencoder captures EEG inter-session variability and thus improves classification accuracy. It compares the performance of a two-classes LDA classifier with and without QVAE denoising on EEG data

Methods

We used an available EEG dataset from 15 participants using 62 electrodes [2].

Preprocessing

- FIR filtering (1-40 Hz), rejection of high-amplitude channels, ICA, ICLabel IC rejection, re-referencing.
- Downsampling (128 Hz), additional IIR filter (1-35 Hz).
- Epoching (64 samples) and xDAWN spatial filtering (4 filters x 2 classes = 8 components) [3].

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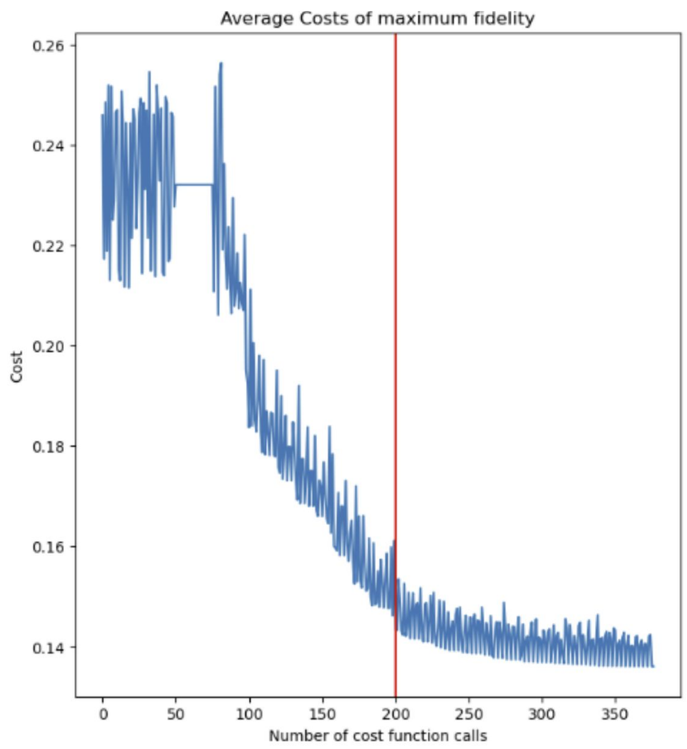
Quantum Denoising

- Quantum autoencoder with a **simulated** 9-qubit (29 = 64 x 8) quantum circuit for encoding.
- The latent space contained 8 qubits.
- A SWAP test [4] was used to measure the fidelity between the original and encoded states.
- Optimized fidelity loss using SPSA [5, 6] with 100 iterations.

Feature Extraction

- Converted epochs to symmetric positive definite matrices, vectorized into 2080-component vectors.
- Classified with LDA, assessing performance with 5-fold cross-validation.

Results



Pipeline	AUC
LDA	0.58
LDA+QVAE	0.51

Pipeline	Time(s)
LDA	0.33
LDA+QVAE	11029.03

Discussion

This study presents an original approach to using QVAE for EEG signal denoising. The results indicated that **quantum denoising did not improve feature extraction**. Several limitations were identified:

- Dimensionality Reduction: Only 8 out of 64 xDAWN components were used, potentially losing critical information. The resultant small QVAE latent space (8 qubits) limited the autoencoder's capacity to encode complex features.
- Algorithm Suitability: The xDAWN filter may not be ideal for this application. Future applications could use a filter bank common spatial patterns approach which reduces dimensionality based on class separability.

This study lays the foundation for further exploration of quantum computing in EEG signal processing, highlighting the need for **larger latent spaces and alternative spatial filtering techniques**.

Although the result are negative, **quantum computing** is **impressively mature enough** to write a Python script for quantum machine learning on a simulated or real quantum computer.

References

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